

REZA TALANEH

Mechanical Engineer | Product Development

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SUMMARY

Mechanical Engineer with more than ten years of experience in industrial product development, manufacturing improvement, and engineering leadership within the polymer film manufacturing industry. Experienced in mechanical design, product validation, design for manufacturing, and cross-functional engineering collaboration. Strong interest in structured product development methodologies, engineering performance indicators, and data-informed engineering decision-making. Seeking doctoral research in product development, production systems, and sustainable manufacturing.

EDUCATION

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| BABOL NOSHIRVANI UNIVERSITY OF TECHNOLOGY (THE Eng. Rank 2026: 501–600) | Babol, Mazandaran, Iran |
| MSc in Applied Mechanical Engineering | Sep 2013 - Feb 2016 |
| <ul style="list-style-type: none">• Attained a GPA of 18.45/20, ranking 2nd among 24 students in the program.• Thesis Title: "Analytical and finite element study of super elasticity behavior and shape memory effect of a Shape memory alloy thick-walled cylinder" | |
| BSc in Mechanical Engineering | Sep 2009 - Sep 2013 |
| <ul style="list-style-type: none">• Achieved a GPA of 17.33/20, securing the 2nd position out of 63 students in the program.• Thesis Title: "Analysis of a sandwich beam with a viscoelastic core using ANSYS" | |

PUBLICATIONS

- Talaneh, R., et al. **Analytical and Finite Element Study of Superelasticity Behavior and Shape Memory Effect of SMA Thin-Walled Cylinders under Torsional Loading**. Proceedings of the 23rd International Mechanical Engineering Conference (ISME2015), Tehran, Iran.
- Talaneh, R., et al. **Analytical and Numerical Modeling of Pseudoelastic and Shape Memory Behavior of Shape Memory Alloys under Uniaxial Tension**. First National Conference on Applied Mechanics, Tabriz, Iran.

RESEARCH INTERESTS

- Product and Production Development
- Sustainable Product Realisation
- Engineering Performance Indicators
- Product Development Methodologies
- Data-Informed Decision-Making
- Design for Manufacturing (DFM)
- Finite Element Analysis

TEACHING EXPERIENCE

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| BABOL NOSHIRVANI UNIVERSITY OF TECHNOLOGY | Babol, Mazandaran, Iran |
| Teaching Assistant – Mechanics of Materials Laboratory | FEB 2014 - Feb 2015 |
| <ul style="list-style-type: none">• Assisted in delivering more than 20 undergraduate Mechanics of Materials laboratory sessions, supporting experimental instruction and student learning. | |

PROFESSIONAL EXPERIENCE

Industrial experience focused on product development, manufacturing improvement, and engineering leadership within the polymer film industry.

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| ABCHECK CO (Manufacturer of Agricultural & Industrial Plastic Blown Films) | Mahmudabad, Mazandaran, Iran |
| Engineering Manager | Oct 2017 – Present |
| <ul style="list-style-type: none">• Led multidisciplinary engineering teams responsible for product development and manufacturing improvement of polymer film products. | |

- Directed engineering initiatives to improve product performance, manufacturing efficiency, and engineering processes.
- Coordinated design, production, quality, and management teams while monitoring engineering KPIs and project outcomes.
- Supported engineering decision-making across product development projects through performance monitoring and cross-functional collaboration.

Design and Development Manager

Sep 2016 – Sep 2017

- Led engineering design, prototype development, testing, and validation of polymer film products and manufacturing solutions.
- Coordinated cross-functional product development activities from concept to implementation.
- Evaluated customer requirements and market opportunities to support engineering innovation.

Research and Development Supervisor

Sep 2013– Sep 2016

- Supervised engineering activities for new product development and manufacturing process improvement.
- Supported tooling, process optimization, product validation, and quality improvement initiatives.
- Contributed to engineering solutions for improving productivity and product performance.

SELECTED MECHANICAL DESIGN PROJECTS

- **Blown Film Dual-Lip Air Ring:** Reverse-engineered and modeled an industrial dual-lip air ring used in blown film extrusion, emphasizing manufacturability and mechanical functionality.
- **Two-Stage Parallel-Shaft Gearbox Design:** Complete gearbox design including transmission calculations, shaft design, bearings selection, housing development, motion verification, and manufacturing drawings.
- **Blueprint-to-CAD Reconstruction & Structural FEA:** Reconstructed production-ready CAD models from engineering drawings and performed structural FEA to evaluate stress distribution and mechanical behavior.
- **Sheet Metal Enclosure:** Designed a manufacturing-ready enclosure applying sheet metal design principles, DFM, forming tools, ventilation design, and fabrication workflows.

CORE COMPETENCIES

Engineering

- Product Development • Mechanical Design • CAD • DFM • FEA
- Engineering KPIs • Data-Informed Decision-Making
- Cross-Functional Collaboration • Engineering Leadership

Methods

- Agile • Scrum • Lean

Software

- SolidWorks • ANSYS • MATLAB • Power BI • Azure DevOps

LANGUAGES

- English (PTE Academic: **Overall 90**)
- Persian (Native)

PROFESSIONAL MEMBERSHIP

- International Association of Engineers (IAENG)

INTERESTS

Guitar • Reading • Cooking • Hiking